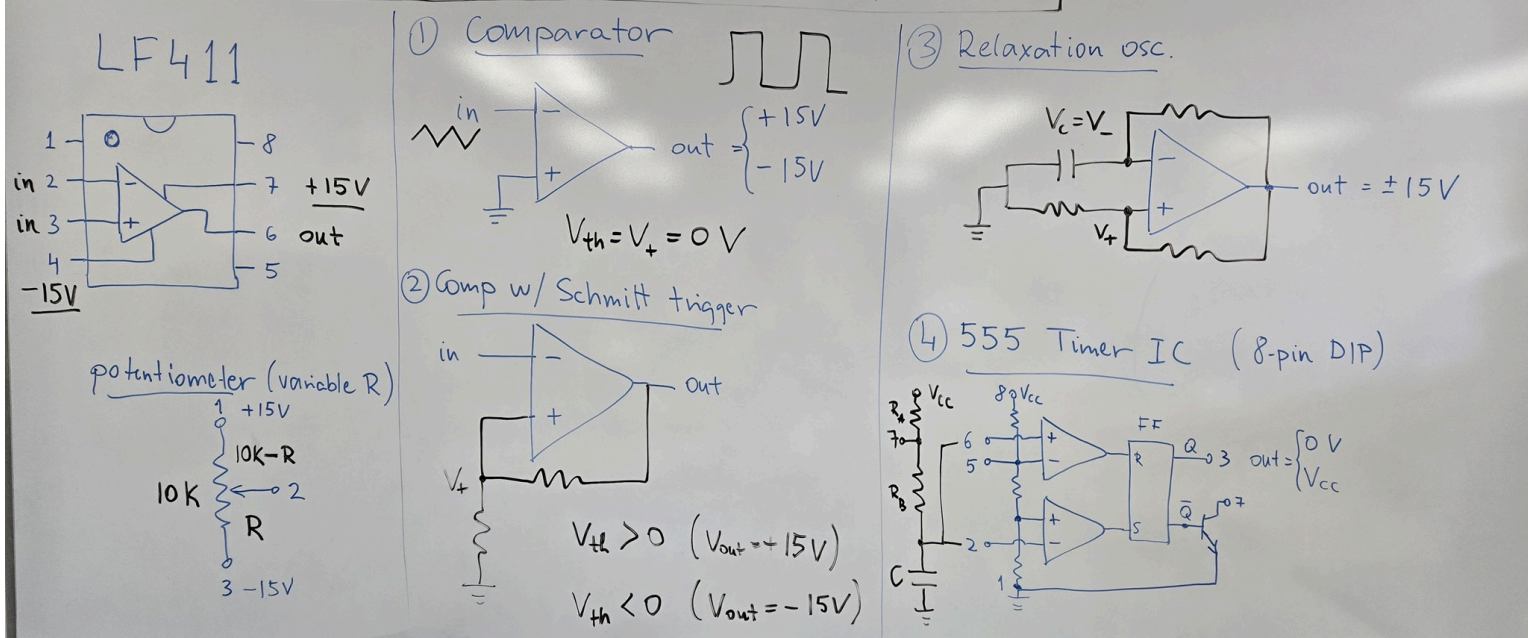


Physics 0525: Lab 9

Op-amps: positive feedback (Comparator, Schmitt trigger, Oscillator)

Equipment: Breadboard, function generator, oscilloscope, resistors, capacitors, decade resistor

Lab 9: Op-amps II (Comparator, oscillator, timing)



Part 1: Comparator

An open loop op-amp can be used as a comparator, a device which outputs one voltage if the non-inverting (+) input voltage is higher than the inverting (-) input voltage, and a different voltage if the opposite is true. Due to the huge open-loop gain of the op-amp, even a miniscule difference between the two input voltages drives the output to either the minimum or maximum output voltage. You have already seen this behavior in last week's lab.

Because of the extremely high open-loop gain, comparators are susceptible to noise when the input voltage difference is near zero. You will observe the effect of "simulated noise" on a simple comparator circuit. Later, you will create a Schmitt trigger using positive feedback to add hysteresis and decrease the number of "false triggers".

Build the summing amplifier and comparator circuit shown in Figure 1 with two LF411 op-amps. The summing amplifier combines the signal with simulated noise, which is used as the input to the comparator. The signal is a triangular wave with ~200 Hz frequency and ~2Vpp amplitude generated by the function generator built into the breadboard. The simulated noise is a

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square wave with 10 kHz frequency and ~ 0.1 Vpp amplitude from the standalone function generator.

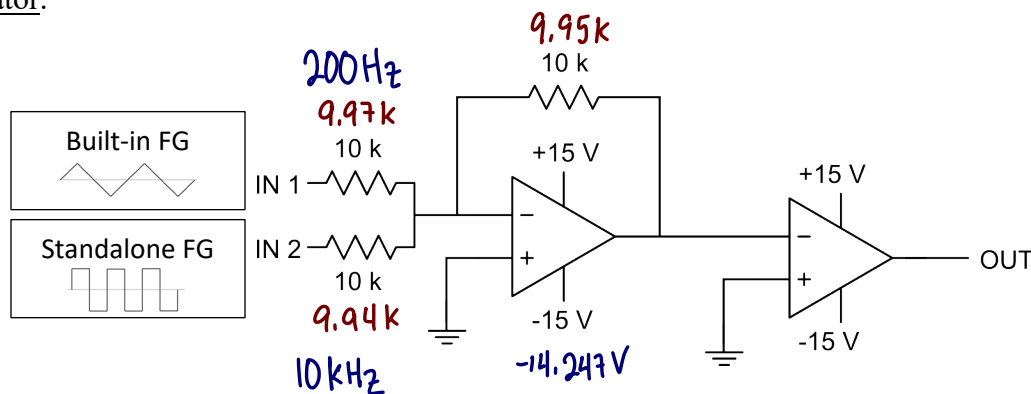
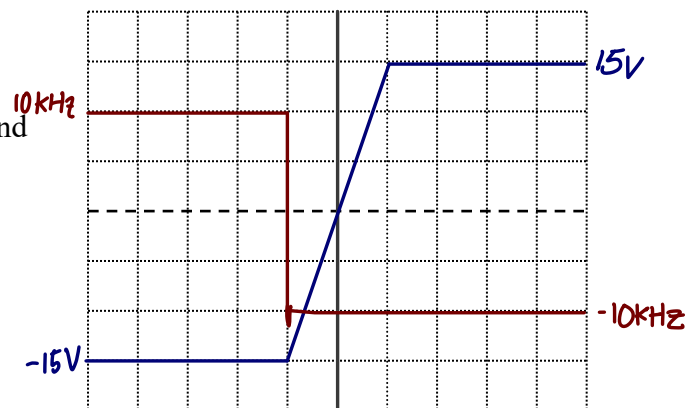


Figure 1: Summing amplifier and comparator.

Display the signals at the inverting – input and the output of the comparator (second op-amp) on your oscilloscope at the same time. First, observe the comparator input and output without noise added and sketch the input and output waveforms:



Describe your observation. When does the output change polarity? Why is the threshold around 0V?

It starts to change when the input flips polarity and the magnitude flips.

Threshold is 0V because the comparator is comparing to ground
i.e. the non-inverting input is grounded

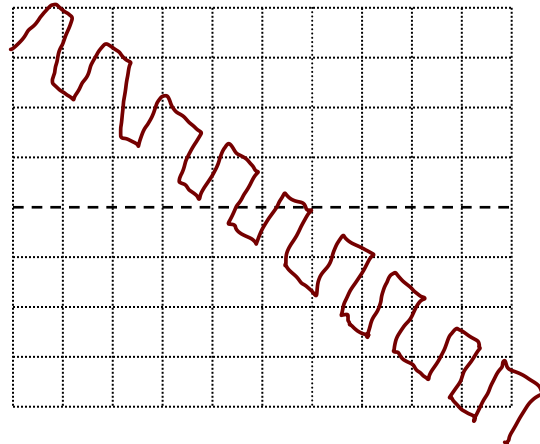
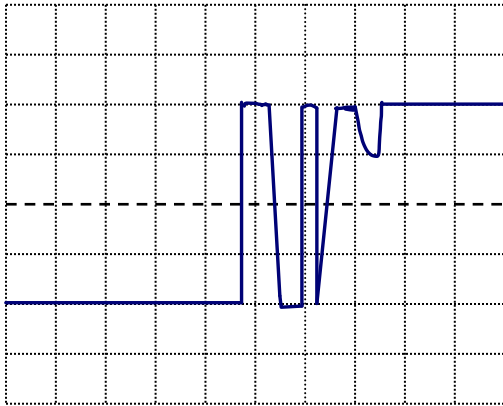
Now add noise to the signal and gradually increase the amplitude of the noise up to 1 Vpp. What is the effect of the added noise on the output of the comparator? Explain. (Look at the rising or falling edge of the output closely.)

It flips many more times as the noise causes the input to cross the threshold many more times

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Sketch the input and output waveforms with enough noise added to see the effect on the output:



The oscilloscopes can measure the transition rate (trigger rate) of the signal which is displayed in the lower right corner of the screen.

Make sure the scope is triggered by the channel corresponding to the comparator input signal and set the scope trigger level near 0V. Measure the transition (trigger) rate of the ~200 Hz input signal without noise added.

_____ 200 _____ [Hz]

Change the scope trigger source to the channel corresponding to the output signal and set the scope trigger level near 0V. Now measure the transition (trigger) rate of the output of the comparator circuit without noise added:

_____ 200 _____ [Hz]

Now measure the transition (trigger) rate of the output of the comparator circuit with noise added (set the noise amplitude to 200 mV):

_____ 910 _____ [Hz]

Do they match? Why or why not? How does the measured frequency depend on the simulated noise amplitude?

NO, they do not match - the measured frequency is dependent on the frequency which the threshold is crossed; since the noise causes multiple zero crossings, the comparator output can toggle several times - as noise amplitude increases, the measured trigger rate increases.

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Part 2. Comparator with Schmitt trigger

Next, you will add positive feedback to the comparator and create a Schmitt trigger. Schmitt-trigger inputs have the advantages that they operate reliably from slow edges and have high immunity to noise. Modify your circuit as shown in Figure 2, keeping the summing amplifier and its inputs the same as before. Use a decade resistor box to set R (between 0-100 k Ω) in the comparator circuit to allow easy variation of the feedback magnitude. Do not forget to remove the wire connecting the op-amp + input to ground in the previous setup!

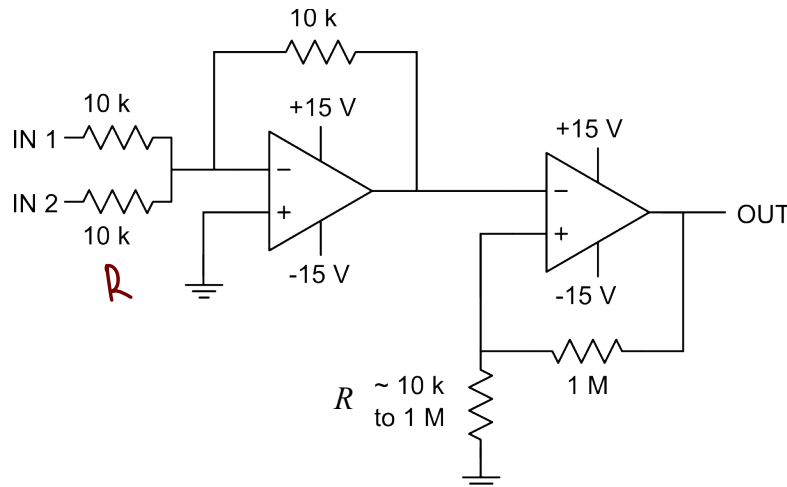


Figure 2: Summing amplifier and comparator with Schmitt trigger.

Observe the signals at the inverting – input and the output of the comparator at the same time. First, display input and output without noise added. You should observe a transition of the output from low state (-15V) to high state (+15V) or vice versa when the input falls or rises above a certain threshold. You should note that the turn-on threshold is below zero and the turn-off threshold is above zero, and the separation between the two thresholds is increasing with increasing R.

The comparator threshold is set by the voltage at its non-inverting + input. Do you see why the threshold is different than zero when $R > 0$ and why the threshold changes when the output of the comparator switches sign? What is the expected comparator threshold in terms of R when the output of the comparator is +15V and when the output is -15V?

$$V_T = V_{out} \left(\frac{R}{R + 1M\Omega} \right)$$

R set to 10k Ω

$$\frac{R}{R + 1.009M\Omega} (V_s)$$

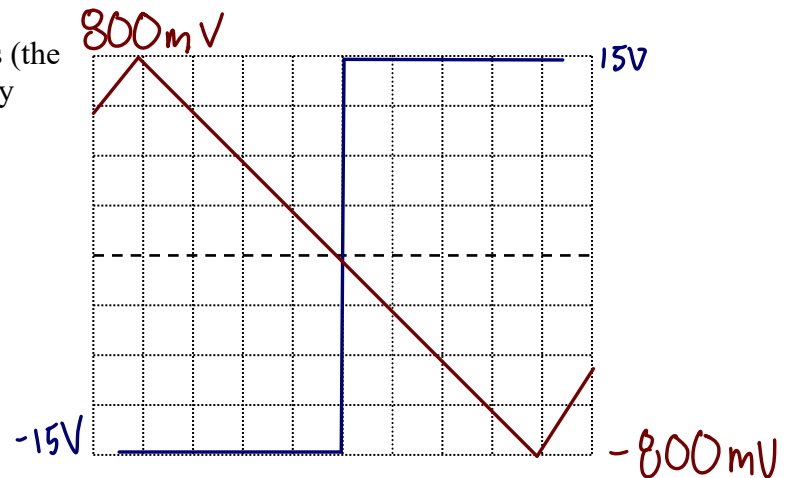
$$\frac{10k}{10k + 1.009M\Omega} \cdot (\pm 15V) \Rightarrow \pm 0.147V$$

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Note that a small (zero) value of the resistor R in the feedback loop reduces the hysteresis, i.e. the difference between the two thresholds, while a large R value increases the hysteresis. This hysteresis ensures that a noisy input wave form does not cause multiple transitions in the output. Depending on the amplitude of the noise, a different threshold is required to eliminate the noise-triggered transitions.

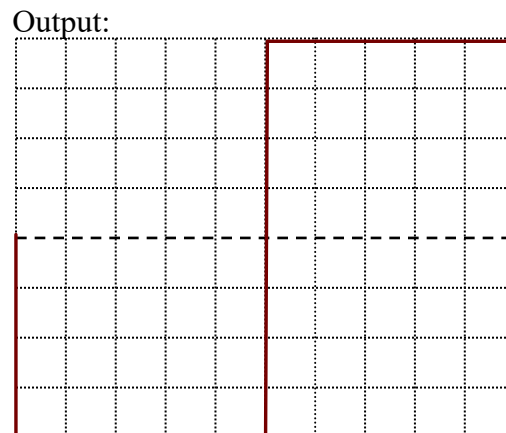
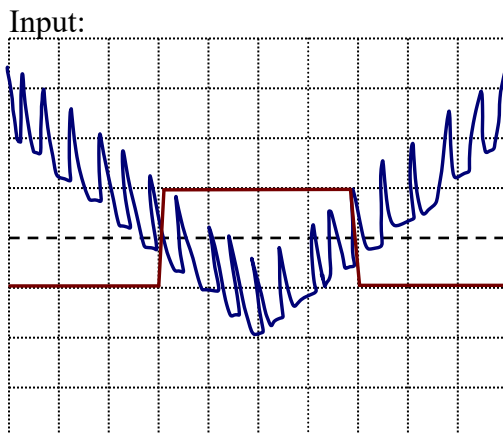
Sketch the input and output when the hysteresis (the difference between the two thresholds) is clearly visible without noise added.



What happens if R (and the threshold) is too high? Why?

If R (and the threshold) is too high then the threshold will not be crossed and no flip will occur on the output

Now add noise to input signal and gradually increase the noise amplitude to 1 Vpp. Notice how R (and the hysteresis) has to increase in order to eliminate the noise induced transitions in the comparator output. Sketch the input and output with just enough hysteresis to eliminate most of the spurious transitions from the noise. Indicate the threshold levels on the input signal by displaying the voltage on the + input of the comparator on the scope (instead of the output).



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In your own words, summarize how positive feedback adds hysteresis and why this can minimize or eliminate the effects of noise in a comparator output.

the output of the first comparator increases the threshold to trigger the threshold of the second's comparative circuit to an extent that bypasses the threshold of the noise. Feedback makes the switching threshold depend on the state where

high output - threshold shifts (+)
low output - threshold shifts (-)

These two states create different switching points instead of one such that small noise near zero does not repeatedly retrigger the comparator.

Measure the transition (trigger) rate with the oscilloscope to quantify the effect of the hysteresis.

First, measure the frequency of the input signal (without noise!): 200 [Hz]

Now measure the transition (trigger) rate of the output of the comparator with noise added and several values of resistor R in the feedback loop and at least two values of the noise amplitude.

Noise amplitude: 200 mVpp

R[Ω]	0	5k	8k	11k	12k	13k	14k	15k
frequency [Hz]	2k	1.4k	1.05k	640	495	350	220	200

Noise amplitude: 500 mVpp

R[Ω]	0	7k	14k	21k	28k	35k	42k	50k
frequency [Hz]	5.25k	4.4k	3.5k	2.5k	1.5k	580	210	200

Part 3. Relaxation oscillator (astable multivibrator)

Although the circuit in Figure 3 has no external input, it will produce an oscillating output. This is achieved by applying both positive and negative feedback. The positive feedback sets the threshold voltage on the non-inverting (+) input just like in the Schmitt trigger. Depending on the output the threshold is either positive (when the output is high +15 V) or negative (when the output is low -15V). The negative feedback is a simple RC circuit with the inverting (–) input being equal to the voltage across the capacitor. Whenever the output changes polarity the capacitor voltage (V_-) starts to climb toward the output voltage with a time constant equal to $\tau = R_t C$. However, when the capacitor voltage (V_-) crosses the threshold voltage (V_+) the output flips sign and the process starts over with the capacitor voltage (V_-) chasing after the new output voltage. Therefore, the output is going to oscillate spontaneously between its two possible states with a fixed 50% duty cycle.

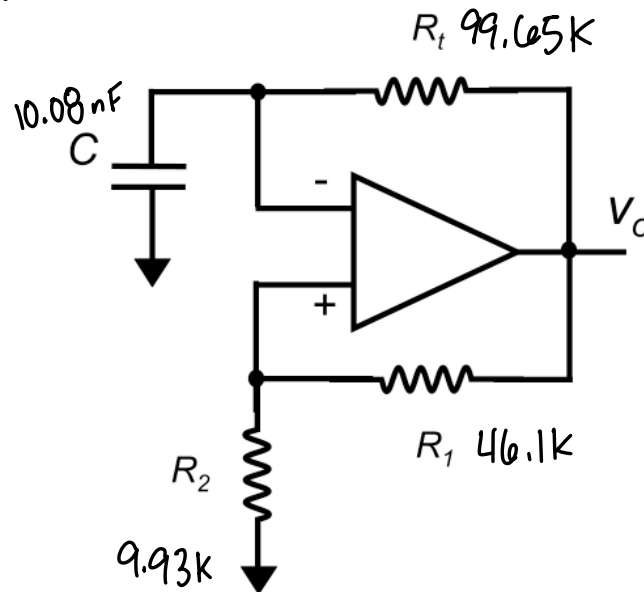
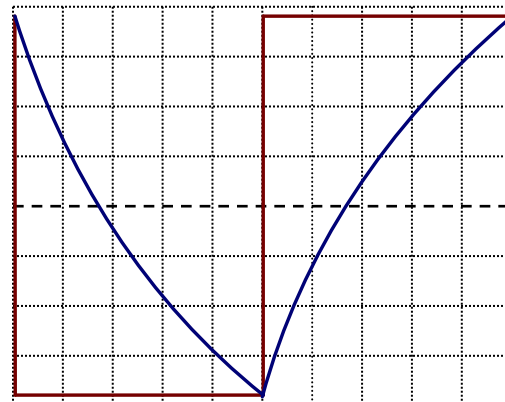


Figure 3. RC relaxation oscillator circuit.

Build the relaxation oscillator using an LF411 op-amp and the following components $R_t = 100\text{ K}\Omega$, $R_1 = 47\text{ K}\Omega$, $R_2 = 10\text{ K}\Omega$, $C = 0.01\text{ }\mu\text{F}$.

Display the voltages at the output and at the inverting input (V_-) on the oscilloscope at the same time. Sketch the waveforms.



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What is the voltage of V- when the output switches polarity low to high and high to low?

low $\Rightarrow$ high : $\approx 2.25V$

high $\Rightarrow$ low : $\approx 2.5V$

Calculate the expected threshold (V+) when the output is high or low based on the positive feedback loop? Do they agree with your measurement above?

$$V_+ = V_{out} \left(\frac{R_2}{R_1 + R_2} \right) \Rightarrow 14.247V \left(\frac{9.93}{46.1 + 9.93} \right) \approx \pm 2.52V$$

Yes! they agree

Measure the period of oscillations. Does it agree with the expected value $T = 2R_t C \ln \frac{R_1 + 2R_2}{R_1}$?

$$f = 1.362 \text{ kHz}$$

$$T_{\text{meas}} = \frac{1}{f} = 7342 \text{ ms}$$

$$T_{\text{calc}} = 7196 \text{ ms}$$

Change $R_t = 20 \text{ K}\Omega$ and measure the period of oscillations now.

$$T = .144 \text{ ms}$$

$$f = \frac{1}{T} = \frac{1}{0.144 \times 10^{-3}} = 6.92 \text{ kHz}$$

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Part 4: 555 Timer Circuit

The 555 IC timer makes it very easy to design moderate frequency oscillators (up to 100 kHz) with precise pulse lengths. The 555 can be used in a monostable one-shot mode, astable oscillator mode, or bistable Schmitt-trigger mode. The internal circuit of the 555 is shown in Figure 4. The timing of the 555 pulses is set by an external timing resistor (R_a) and capacitor (C).

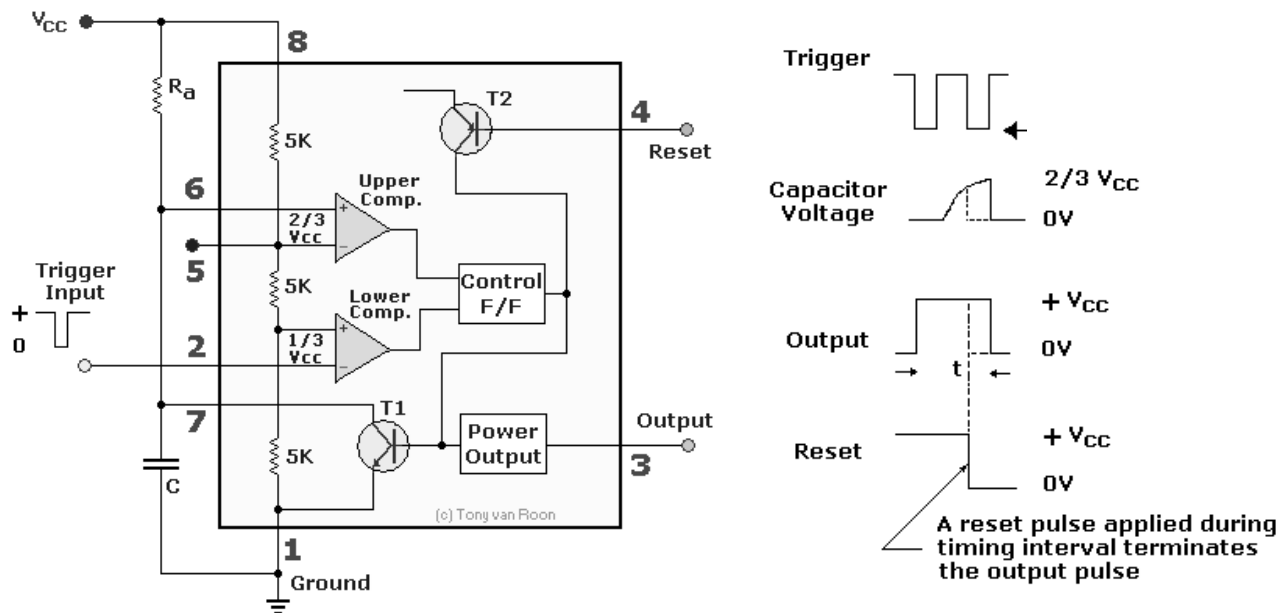


Figure 4: Internal circuitry of the 555 and set up for one-shot operation.

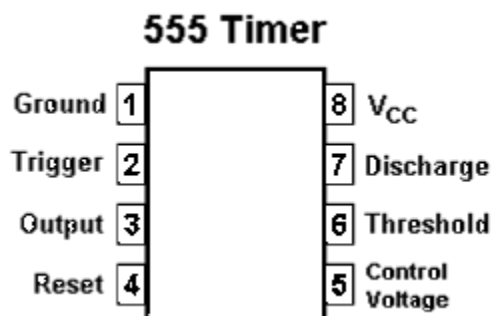


Figure 5: Pin diagram of the 555.

Note: The 555 timer takes pulses that go from 0 to $+V_{CC}$

Do not use square waves with the 555 timer!

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The circuit shown in Figure 6 makes the 555 into a free running (astable) oscillator. The external capacitor C charges through $R_A + R_B$ and discharges through R_B . Thus, the period and the length of the pulses can be set precisely by these two resistors and C . In this mode of operation, the timing capacitor C charges and discharges between $1/3$ and $2/3$ of the supply voltage V_{CC} . The supply voltage can be anything between 5V and 15 V. Build the circuit and test it.

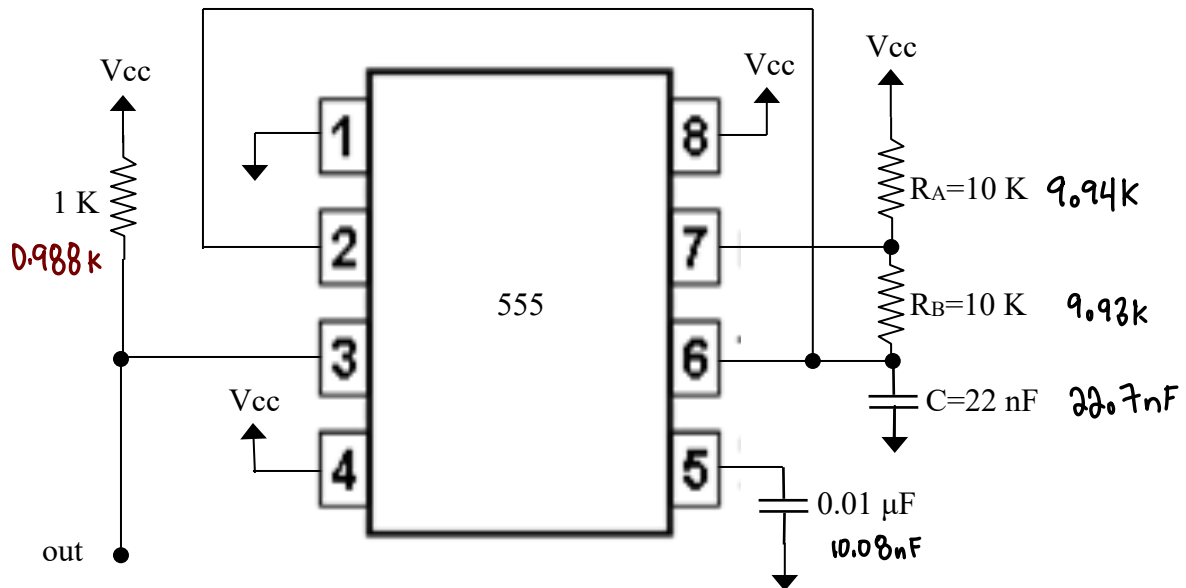


Figure 6: 555 free running (astable) oscillator.

Display the output on the scope and measure the period of the oscillator.

$$T = \underline{0.464} \text{ [ms]}$$

Does it agree with the expected period $T \approx 0.7(R_A + 2R_B)C$?

$$T = \underline{0.474} \text{ [ms]}$$

Measure the length of time the output is high (T_h) during one period. Calculate the duty cycle of the oscillator, i.e. the fraction of the period when the output is high: T_h/T .

$$T_h = 4.64 \quad 31/4.64 = 0.67$$

Does it agree with the expected duty cycle $D = (R_A + R_B)/(R_A + 2R_B)$?

$$= \frac{(9.94 + 9.93)}{[9.94 + 2(9.93)]} \Rightarrow \underline{0.666}$$

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Does the period change with supply voltage between 5-15 V? Make sure you do not exceed the absolute maximum voltage of 18V!

NO change to period only shifts amplitude

Set V_{CC} back to 15 V. See what happens to the period when the control voltage (pin 5) is changed between 3V and V_{CC} ? (Note that the behavior will change when the control voltage becomes too low, so keep it above $\sim 3V$.) Use a 10 k Ω potentiometer as a voltage divider between 0V and V_{CC} to adjust the control voltage.

Control Voltage	Period
0 V	0
4 V	0.22 ms
8 V	0.35 ms
12 V	0.629 ms
15 V	1.38 ms

How might this be useful? For making kickass EDM

or used as a voltage controlled oscillator where the frequency of oscillation can be adjusted electronically for timing/tone/modulation

Dani